

Instructions:

1. Answer all questions. Each question carries 10 marks.
2. Elegant and innovative solutions will get extra marks.
3. Diagrams and justification should be given wherever necessary.
4. Before starting to answer, fill in the FACE SLIP completely.
5. Your 'rough work' should be done in the answer sheet itself.
6. Maximum time allowed is THREE hours.

1. Find real numbers 'a' and 'b' such that

$$(a+b)(a^2-b^2) = 9$$

$$\text{and } (a-b)(a^2+b^2) = 5$$

2. The sum of the digits of a number of two digits is 'x'. If the digits of that number are interchanged, then the value of the original number is increased or decreased by 'y'. Prove that the original number is  $\frac{11x \pm y}{2}$ .
3. ABC is a triangle. AE and BF are perpendiculars drawn from A, B to the internal bisector of  $\angle C$ . The midpoint of AB is G. Prove that the line segment joining G to either E or F is equal to half the difference of the sides AC and BC.
4. Determine all natural numbers 'n' such that  $\frac{(n+1)^2}{n+7}$  is also a natural number.
5. Find all positive integers x, y, z such that  $2x^2y^2 + 2y^2z^2 + 2z^2x^2 - x^4 - y^4 - z^4 = 576$ .
6. If a, b, c are all different and non-zero and if  $\frac{1}{a^3(b-c)} + \frac{1}{b^3(a-c)} = \frac{1}{ab(a-c)(b-c)}$ , prove that  $\frac{1}{a} + \frac{1}{b} = \frac{1}{c}$  or  $a^2 + b^2 = ab$ .
7. If a, b, c, d are positive reals, prove that
 
$$\frac{a^2 + b^2 + c^2}{a+b+c} + \frac{b^2 + c^2 + d^2}{b+c+d} + \frac{c^2 + d^2 + a^2}{c+d+a} + \frac{d^2 + a^2 + b^2}{d+a+b} \geq a+b+c+d$$
8. ABCDEF is a convex hexagon in which the opposite sides are parallel. Also,  $\angle A = \angle D = 90^\circ$ . The distances between the opposite parallel sides, taken in pairs, are also equal. Prove that the diagonals BE and CF include an angle  $45^\circ$ .

End of Question Paper.